Grace Jia

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gjia25.github.io

I work on the security and privacy of cloud and AI systems, focusing on preventing information leakage from AI/ML services, encrypted storage, confidential computing, and distributed analytics. My research exposes such leakage through side-channel attacks, proves theoretical lower bounds on the cost of preventing it, and builds practical defenses. Recently, I am focused on agentic system security.

Areas of Interest: Systems, Security, Applied Cryptography

EDUCATION

Yale University, New Haven, CT *Aug 2022 – May 2027 (Expected)*
Ph.D. in Computer Science
Advisor: Prof. Anurag Khandelwal
Selected courses: Big Data Systems, Distributed Systems, Blockchain and Cryptocurrency

Cornell University, Ithaca, NY *Aug 2018 – May 2022*
B.A. in Computer Science, magna cum laude
GPA: 3.97/4.00
Selected courses: Cryptography, System Security, Computer Networking, Database Systems

GRANTS & AWARDS

- **Yale Kwok Family Scholarship Fund 2024**
- **Yale Student Fellowship 2022**

PUBLICATIONS

[1] Grace Jia, Alex Wong, Anurag Khandelwal. **Found in Translation: A Generative Language Modeling Approach to Memory Access Pattern Attacks.** *USENIX Security Symposium (USENIX Security 2025)*

[2] Mahdi Soleimani, Grace Jia, Anurag Khandelwal. **Weave: Efficient and Expressive Oblivious Analytics at Scale.** *USENIX Symposium on Operating Systems Design and Implementation (OSDI 2025)*

[3] Grace Jia, Rachit Agarwal, Anurag Khandelwal. **Length Leakage in Oblivious Data Access Mechanisms.** *USENIX Security Symposium (USENIX Security 2024)*

PUBLICATIONS (under submission/preparation)

** denotes equal contribution.*

[4] Grace Jia, Landon Cox, Pedro Henrique Penna, Anurag Khandelwal. **Ziploc: Bounding Privacy Leakage from Coding Agent Systems.** *In preparation, 2026*

[5] Mahdi Soleimani*, Grace Jia*, Seung-seob Lee, Anurag Khandelwal. **Hop, Skip, and a Logarithmic Jump: Hidden Costs of Hiding Correlations in Oblivious Data Stores.** *In submission, 2026*

[6] Mahdi Soleimani, Grace Jia, In Gim, Seung-seob Lee, Anurag Khandelwal. **Wiretapping LLMs: Network Side-Channel Attacks on Public LLM Services.** *In submission, 2025*

RESEARCH EXPERIENCE

Microsoft Research Redmond – Security Research Group *June 2026 – Present*
Research Intern & Collaborator
Mentors: Landon Cox, Pedro Henrique Penna

- Designed harness for LLM agents to prevent data exfiltration under prompt-injection attacks
- Implemented budgeting mechanism atop information-flow control to deterministically bound privacy leakage
- Exploring extensions and deployment on GitHub Agentic Workflows

Yale Computer Science Department *Aug 2022 – Present*

Research Assistant

Advisor: Prof. Anurag Khandelwal

- Proved lower bounds in bandwidth–latency–cache tradeoff for hiding correlated access patterns
- Implemented attack on correlated access patterns in TEE memory with 70–99% accuracy
- Formulated network side-channel attack on LLM services with up to 92% accuracy
- Developed Weave system for oblivious cloud analytics, improving execution times by 4–10× over prior state-of-the-art

Cornell Computer Science Department *Feb 2021 – May 2022*

Research Assistant

Advisor: Prof. Rachit Agarwal

- Designed length-hiding oblivious access mechanisms for various leakage scenarios and proved performance lower bounds
- Developed new analytical framework for length leakage setting and security-performance tradeoff

INDUSTRY EXPERIENCE

Palo Alto Networks, Santa Clara, CA *Summer 2021*

Cloud Services Portal Engineer Intern

- Deployed single sign-on feature using JSON Web Tokens to establish shared Identity and Access Management (IAM) system across all company microservices

Klaviyo, Boston, MA *Summer 2020*

Software Engineer Intern

- Scaled up asynchronous task system for profile CSV exports using RabbitMQ and Celery, handling greater customer size and demand

Applied Science and Technology Research Institute, Hong Kong *Summer 2019*

Summer Intern

- Evaluated statistical and deep learning methods for fake news classification

TALKS AND PRESENTATIONS

Mitigating Access Pattern Leakage in Cloud Systems

Boston University *Feb 2026*

Brown University *Feb 2026*

University of Pennsylvania *Mar 2026*

Found in Translation: A Generative Language Modeling Approach to Memory Access Pattern Attacks

USENIX Security, Seattle *Aug 2025*

Length Leakage in Oblivious Data Access Mechanisms

USENIX Security, Philadelphia *Aug 2024*

TEACHING EXPERIENCE

Teaching Fellow in the Computer Science department at Yale University:

CPSC 422: Design & Implementation of Operating Systems *Spring 2024, Spring 2025*

Teaching Assistant in the Computer Science department at Cornell University:

CS 2800: Discrete Structures *Spring 2019, Spring 2020*

COMMUNITY AND PROFESSIONAL SERVICE

Technical Program Committee: Deepfake, Deception, and Disinformation Security Workshop (3D-Sec) 2025

Community Service:

- Graduate Student Assembly (GSA) Representative at Yale University, 2025 – Present
  Elected as Steering Committee Member out of 14 departments in Physical Sciences & Engineering Division
- Computer Science Graduate Student Advisory Committee (CS GSAC) Member at Yale University, 2026